

SUPREETH KUMAR KUMARASWAMY

TECHNICAL PROGRAM MANAGER

Chief Architect – Autonomy, Robotics & AI Systems

+1 (646) 932-3263 | supreethk666@gmail.com | [linkedin.com/in/supreeth-kumar-k](https://www.linkedin.com/in/supreeth-kumar-k) | github.com/supreethk6

PROFESSIONAL SUMMARY

Technical Program Manager and Chief Architect with 9+ years designing and delivering complex autonomous robotics systems from concept through nationwide production deployment. Architect of end-to-end autonomy stacks integrating ROS2/Autoware frameworks, multi-modal sensor fusion pipelines, real-time SLAM algorithms, and AI-powered perception systems. Led cross-functional teams of 15+ engineers delivering safety-critical platforms with 3M+ operational hours and zero safety incidents.

Deep expertise in autonomous navigation architecture, sensor fusion design, and embedded systems integration. Successfully led FedRAMP authorization achieving Authority to Operate on federal networks. Recognized as the principal technical authority and indispensable driving force behind mission-critical robotics programs advancing U.S. technological leadership.

PROGRAM IMPACT METRICS

100+ Robots Deployed	60% Build Time Cut	53% Incidents Reduced	3M+ Patrol Hours
\$20M+ Revenue	15+ Engineers Led	FedRAMP Authorized	Zero Safety Incidents

PROFESSIONAL EXPERIENCE

KNIGHTSCOPE, INC. – Mountain View, CA

Lead Autonomy and Robotics Engineer & K7 Technical Program Manager | Aug 2019 – Present

Chief Architect – Autonomy, Robotics & AI Systems

K7 AUTONOMOUS SECURITY ROBOT – Chief Architect & Technical Program Manager

- Architecting and leading the most advanced outdoor autonomous security platform, a 4-wheeled multi-terrain robot for military bases, airports, and critical infrastructure.*
- Autonomy Stack Architecture:** Designed complete ROS2 Humble autonomy stack, implemented NDT (Normal Distributions Transform) scan matching for less than 10cm localization accuracy, Lanelet2 HD map integration, hybrid A* path planner with Reeds-Shepp curves, and MPC (Model Predictive Control) trajectory tracking with 50ms control loops
 - Sensor Fusion Pipeline:** Architected multi-modal perception system, Velodyne VLP-16 LiDAR (1.2M points/sec), Continental ARS408 radar, FLIR thermal camera, IMU/GPS/VO (visual odometry) fusion using Extended Kalman Filter; implemented point cloud preprocessing (voxel grid filtering, ground plane segmentation, Euclidean clustering) achieving 360° obstacle detection at 20Hz
 - Real-Time Systems:** Configured PREEMPT_RT Linux kernel for deterministic control; designed CAN bus communication architecture interfacing with vehicle control unit; implemented safety watchdog with hardware heartbeat monitoring and fail-safe stop behaviors
 - AI Perception Systems:** Integrated RealSense D435i depth cameras; Integrated TensorRT-optimized deep learning models on NVIDIA Jetson Orin, YOLOv8 object detection (persons, vehicles, weapons), DeepSORT multi-object tracking, acoustic CNN for gunshot/glass break classification; achieved <100ms inference latency
 - Program Execution:** Led 12+ engineer cross-functional team through concept→alpha prototype→beta using Agile/Scrum; managed external chassis vendor integration; delivered Board presentations and DoD/DHS demonstrations contributing to U.S. Air Force SBIR contract

K5 5TH GENERATION ROBOT – Chief Architect & Technical Program Manager

Architected and managed complete product lifecycle for flagship 2-wheeled balancing security robot deployed from NYPD Times Square to VA Medical Centers nationwide.

- **Navigation Stack Redesign:** Led ROS1→ROS2 migration; implemented AMCL (Adaptive Monte Carlo Localization) with particle filter optimization, DWA (Dynamic Window Approach) local planner, costmap2d with inflation layers; integrated Vedlodyne VLP-16 LiDAR achieving ±5cm positioning accuracy indoors
- **SLAM Implementation:** Deployed Gmapping and Cartographer SLAM for real-time map building; implemented loop closure detection, scan-to-map matching, and pose graph optimization; created map management system for multi-floor deployments with automatic map switching based on elevator detection
- **Balancing Control System:** Tuned cascaded PID controllers for self-balancing platform, inner loop (1kHz IMU feedback), outer loop (100Hz velocity control); implemented sensor fusion combining IMU with wheel encoders using complementary filter
- **Fleet Telemetry & Diagnostics:** Architected MQTT-based telemetry pipeline streaming 50+ sensor parameters at 10Hz to AWS IoT Core; built Grafana dashboards for real-time fleet monitoring; implemented predictive maintenance using motor current anomaly detection, reduced incidents 53%
- **Computer Vision Pipeline:** implemented OpenCV-based license plate recognition (ALPR), face detection using dlib HOG+SVM, and thermal anomaly detection with FLIR Lepton 3.5 microbolometer
- **Production Scale:** Managed full lifecycle from design→production→deployment; replaced 100% of legacy fleet; achieved 100+ robots deployed, 3M+ operational hours, \$20M+ revenue, 4-year client retention

FEDRAMP AUTHORIZATION PROGRAM – Technical Program Manager

- **Security Architecture:** Designed end-to-end encrypted communication using TLS 1.3 mutual authentication; implemented AES-256 encryption for data-at-rest; architected secure boot chain with TPM 2.0 hardware root of trust
- **Cloud Infrastructure:** Led migration to FedRAMP-authorized AWS GovCloud; implemented VPC isolation, IAM least-privilege policies, CloudTrail audit logging, and GuardDuty threat detection; achieved continuous ATO monitoring
- **Milestone Achievement:** First autonomous security robot to achieve FedRAMP Authority to Operate, enabled "K5 GOV" deployment at Audie L. Murphy VA Medical Center; opened federal/DoD market

ICM & ACM MODULAR IOT PLATFORMS – Chief Architect & Technical Program Manager

- **System Architecture:** Designed modular edge computing platform using NVIDIA Jetson Orin; implemented containerized microservices (Docker) with Kubernetes orchestration for OTA updates; created hardware abstraction layer enabling plug-and-play sensor integration
- **Communication Stack:** Implemented multi-protocol support; designed store-and-forward queue for intermittent connectivity; achieved 99.9% message delivery
- **AI Edge Inference:** Deployed quantized TensorFlow Lite models for on-device inference; implemented INT8 quantization reducing model size 4x while maintaining 95%+ accuracy; enabled real-time threat detection without cloud dependency

AUTONOMOUS CHARGING SYSTEM – Chief Technical Architect

- **Precision Docking Algorithm:** Designed multi-phase docking sequence, coarse approach using LiDAR-based dock detection (RANSAC plane fitting), fine alignment with AprilTag fiducial markers (sub-centimeter accuracy), final insertion using LiDAR, IR proximity sensors and contact detection
- **Power Management:** Implemented intelligent charging scheduler optimizing patrol coverage vs. battery state; designed BMS (Battery Management System) integration monitoring cell voltages, temperatures, and SoC; achieved 24/7 autonomous operation with <5% downtime

COLUMBUS AI – Columbus, OH

R&D Engineer – Autonomous Vehicle Safety Systems | Dec 2017 – Jan 2019

- **AEB System Architecture:** Led development of Autonomous Emergency Braking using sensor fusion, Velodyne VLP-16 LiDAR point cloud processing, Delphi ESR radar tracking, Mobileye camera integration; implemented Time-to-Collision (TTC) calculation with 100ms decision latency
- **Vehicle Integration:** Designed CAN bus interface using Vector CANalyzer/CANoe; implemented J1939/OBD-II protocol parsing; developed brake-by-wire actuation through vehicle gateway; conducted FMEA (Failure Mode and Effects Analysis) for ASIL-B compliance
- **Deep Learning Pipeline:** Developed custom object detection using Faster R-CNN trained on synthetic data from GTA V game engine (domain randomization); implemented data augmentation pipeline generating 100K+ labeled images; achieved 92% mAP on KITTI benchmark
- **Driver Monitoring System:** Built drowsiness detection using dlib facial landmark localization (68-point model); implemented Eye Aspect Ratio (EAR) algorithm for blink detection; trained LSTM network on temporal patterns achieving 96% drowsiness classification accuracy
- **Data Acquisition:** Managed on-road testing program with Honda CR-V and Kia Soul test vehicles; designed ROS bag recording pipeline capturing synchronized sensor data at 2GB/min; built annotation tools for ground truth labeling

TECHNICAL EXPERTISE

Autonomous Systems & Robotics

ROS/ROS2 (Humble, Foxy, Noetic) | Autoware.Universe | SLAM (Cartographer, Gmapping, ORB-SLAM2, LIO-SAM) | Localization (AMCL, NDT, ICP) | Path Planning (A*, RRT*, Hybrid A*, Lattice) | Motion Control (MPC, PID, Pure Pursuit) | Sensor Fusion (EKF, UKF, Particle Filters) | Behavior Trees | State Machines | Nav2 | MoveIt2

Perception & AI/ML

LiDAR Processing (PCL, Open3D) | Point Cloud Segmentation | 3D Object Detection | Camera-LiDAR Calibration | Camera-LiDAR Calibration | Sensor Fusion | Computer Vision (OpenCV) | Deep Learning (PyTorch, TensorFlow) | YOLO | Faster R-CNN | DeepSORT | TensorRT Optimization | NVIDIA Jetson (Nano, Xavier, Orin) | RealSense | FLIR Thermal

Embedded Systems & Hardware

CAN Bus (J1939, CANopen) | Serial Protocols (UART, SPI, I2C) | Real-Time Linux (PREEMPT_RT) | Microcontrollers (STM32, Arduino) | IMU Integration | GPS/RTK | Motor Controllers | BMS Integration | PCB Design | Soldering | Oscilloscope Debugging | Vector CANalyzer/CANoe

Software & Cloud Infrastructure

C/C++ | Python | MATLAB/Simulink | Linux | Docker | Kubernetes | AWS (IoT Core, GovCloud) | MQTT | REST APIs | Git | CI/CD | Jenkins | Ansible | Terraform | FedRAMP | Grafana

Program Management

End-to-End Program Execution | Technical Architecture | Scope/Schedule/Budget | Risk Management | Vendor Management | Cross-Functional Leadership (15+ engineers) | Agile/Scrum | JIRA | Confluence | Executive Communication | Board Presentations | Stakeholder Alignment | SDLC | OKRs/KPIs

EDUCATION

NEW YORK UNIVERSITY – Tandon School of Engineering

Master of Science in Mechatronics, Robotics & Automation Engineering | 2017 – 2019

- **Coursework:** Advanced Mechatronics, Mobile Robotics, Robot Perception, Swarm Robotics, Simulation Tools, Entrepreneurship
- **Research:** Mobile 3D Printing Robots, contributed to \$1.2M NSF grant (Award #1932187)

THOUGHT LEADERSHIP & TECHNICAL MENTORSHIP

CARNEGIE MELLON UNIVERSITY – Robotics Institute

Technical Mentor & Architecture Advisor – MRSD Capstone Project | 2025 – Current

Project: "PatrolKnight: AI-Augmented Patrol Scheduler for Autonomous Security Robots"

- Guiding ROS2/Autoware.Universe integration architecture advising on node composition, lifecycle management, QoS policies, and inter-process communication patterns
- Advising on multi-robot coordination algorithms, fleet task allocation, conflict-free path planning, V2V communication protocols
- Sharing K7 development best practices including simulation-first development, hardware-in-loop testing, and field deployment validation

NEW YORK UNIVERSITY – K-12 STEM Outreach

Graduate Teaching Assistant – NSF Innovative Technology Experiences Program (DRL-1614085)

- Developed drone curriculum teaching flight dynamics, PID control, and computer vision to high school students
- Mentored teachers and students in robotics and entrepreneurship fundamentals, Arduino programming, Fusion-360 CAD, 3D printing and sensor integration

ACADEMIC RESEARCH & TECHNICAL PROJECTS

- **Mobile 3D Concrete Printing (NSF-funded):** Designed multi-robot coordination system using ROS for autonomous construction; implemented visual servoing with ArUco markers for sub-millimeter print head positioning; developed custom G-code interpreter for concrete extrusion control; contributed to \$1.2M NSF grant and ASCE CRC2020 publication
- **Swarm Robotics Simulation:** Implemented Reynolds flocking algorithm and potential field navigation in PyBullet physics engine; developed consensus-based formation control for differential drive robots; achieved stable 10-robot formations with collision avoidance
- **Pharm'automate – Autonomous Pharmacy Robot:** Built complete pick-and-place system using custom end effector; implemented color-based medicine detection using HSV segmentation; designed mobile base navigation using ROS move_base; integrated prescription parsing from QR codes
- **Smart Irrigation System (IoT Startup Project):** Designed sensor network measuring soil moisture, water level, and weather conditions; implemented mesh networking for long-range farm coverage; built React Native mobile app with Firebase backend for remote monitoring and valve control
- **F.L.Y.-m – Phone Drop Protection:** Prototyped mini quadcopter phone case with IMU-based free-fall detection; designed thrust vectoring system using brushless motors; implemented emergency deployment algorithm triggering at 0.3g threshold – received NYU Makerspace prototype funding

STRATEGIC PROGRAM IMPACT

Augmenting Law Enforcement & Public Safety

Systems I've architected now patrol malls, hospitals, corporate campuses, and NYC's busiest subway station. Deployed robots operates at ~\$9/hour providing 24/7 AI-powered surveillance, a force-multiplying deterrent extending police capabilities where human resources are constrained.

Protecting Critical Infrastructure & Defense Assets

K7 platform explicitly designed for military bases, airports, and power installations. U.S. Air Force selected program for SBIR contract to reimagine base security, autonomous perimeter patrol reducing risk to human guards while providing persistent surveillance with AI threat detection.

Advancing U.S. Technological Leadership

Ensuring U.S. independence from foreign robotics technologies for critical security applications. Led FedRAMP authorization, first autonomous security robot achieving Authority to Operate on federal networks, enabling government agencies to deploy domestically-developed AI-powered security solutions meeting stringent federal cybersecurity standards.